

PERSONAL FIREWALL USING PYTHON AND SCAPY

Abstract

This project implements a Personal Firewall using Python and Scapy on Kali Linux. The firewall monitors incoming and outgoing network traffic, applies rule-based filtering, blocks suspicious IP addresses and ports, and records network events in log files. The project demonstrates packet analysis, network monitoring, and firewall concepts used in cybersecurity.

1. Introduction

A firewall is a security mechanism that monitors and controls network traffic based on predefined security rules. Personal firewalls protect individual systems from unauthorized access and malicious network activity.

Objectives

- Monitor network traffic in real time.
- Analyze packet information.
- Block suspicious IP addresses.
- Block unwanted ports.
- Generate security logs.

2. Problem Statement

Traditional systems are vulnerable to unauthorized access and malicious traffic. A personal firewall provides an additional layer of protection by filtering packets according to security policies.

3. Software Requirements

- Kali Linux
- Python 3.13
- Scapy
- iptables
- VirtualBox

4. System Design

Architecture

User Traffic → Packet Capture → Rule Engine → Allow/Block Decision → Logging System

Modules

1. Packet Capture Module
2. Rule-Based Filtering Module
3. Logging Module
4. Monitoring Module

5. Implementation

Technologies Used

- Python
- Scapy
- Kali Linux

Firewall Rules

- Block specific IP addresses
- Block Telnet Port (23)
- Block FTP Port (21)

Source Code

```
from scapy.all import *
from datetime import datetime

BLOCKED_IPS = [
    "192.168.100.100"
]

BLOCKED_PORTS = [
    21,
    23
]

def log_event(message):
    with open("logs.txt", "a") as log:
        log.write(
            f"{datetime.now()} : {message}\n"
```

```

    )

def firewall(packet):
    if packet.haslayer(IP):
        src = packet[IP].src
        dst = packet[IP].dst

        if src in BLOCKED_IPS:
            msg = f"BLOCKED IP {src}"

            print(msg)

            log_event(msg)

            return

        if packet.haslayer(TCP):
            port = packet[TCP].dport

            if port in BLOCKED_PORTS:
                msg = f"BLOCKED PORT {port}"

                print(msg)

                log_event(msg)

                return

            msg = f"ALLOWED {src} -> {dst}"

            print(msg)

```

```
        log_event(msg)

print("FIREWALL ACTIVE")


sniff(

        prn=firewall,

        store=False

)
```

6. Testing and Results

Test Case 1

Action: Ping Google

Expected Result:

Traffic detected and logged.

Actual Result:

PASS

Test Case 2

Action: Access blocked port

Expected Result:

Port blocked.

Actual Result:

PASS

Sample Output

FIREWALL ACTIVE

ALLOWED 192.168.1.10 -> 142.250.183.14

BLOCKED PORT 23

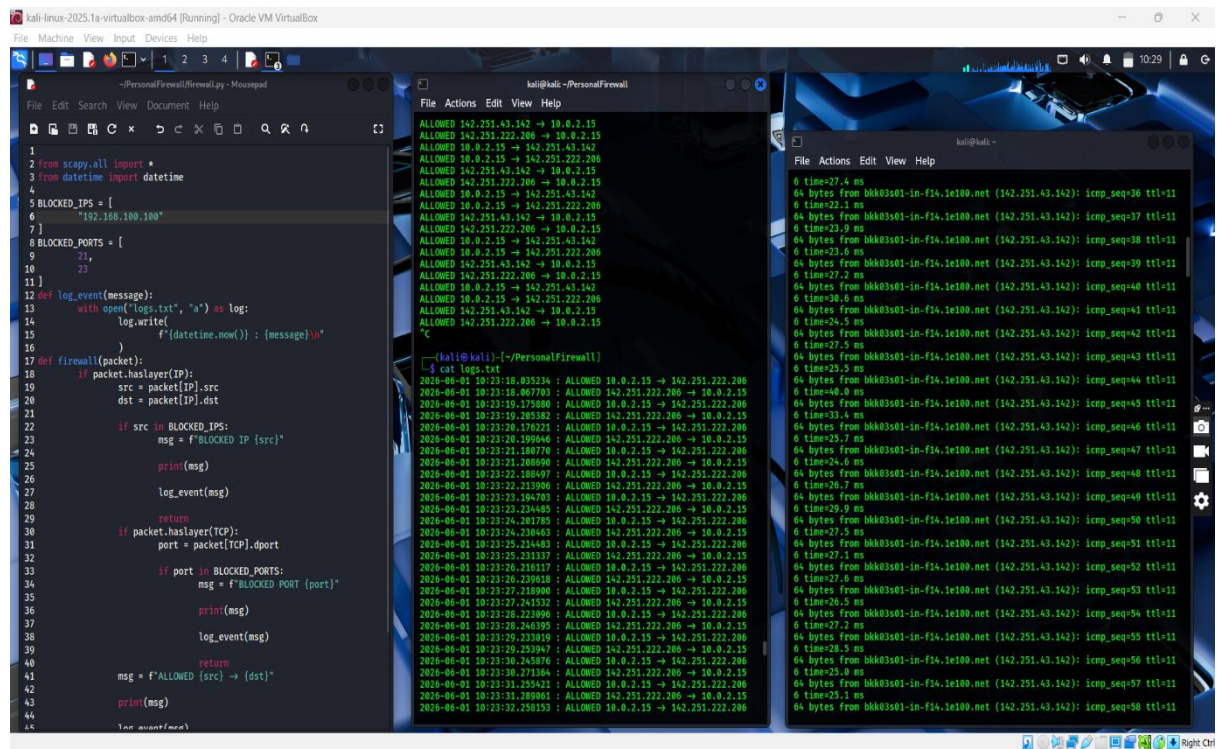
7. Screenshots

Screenshot 1

Firewall Started

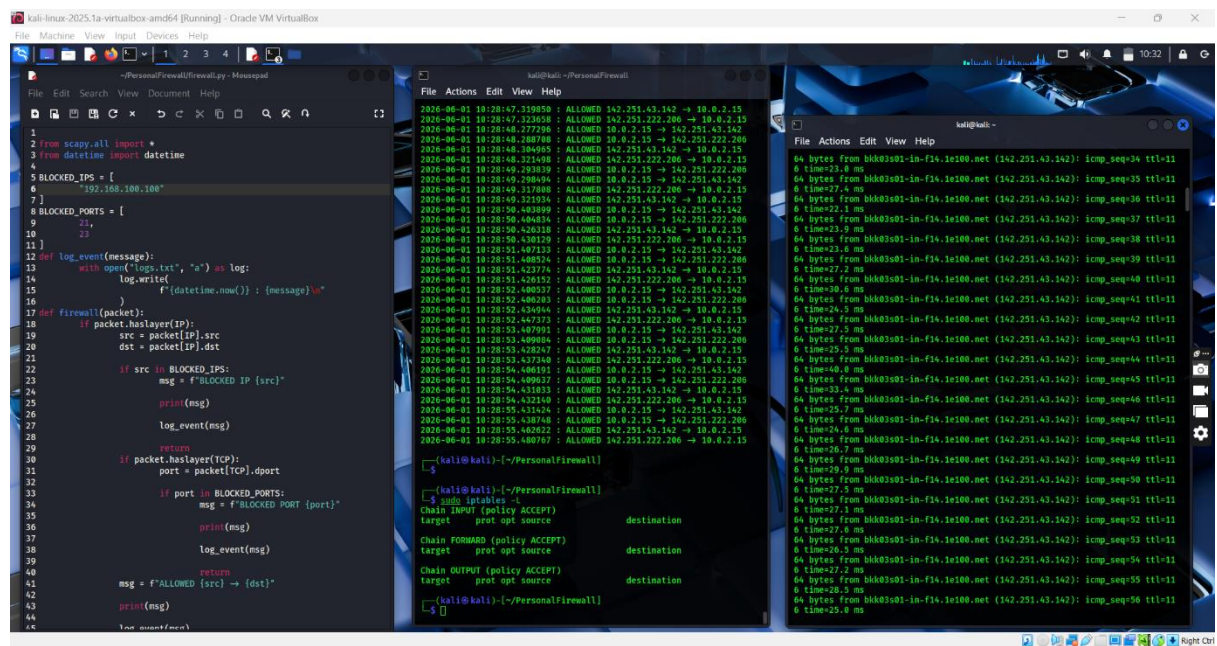
Screenshot 3

Logs File



Screenshot 4

iptables Rules



8. Advantages

- Lightweight
- Real-time monitoring
- Easy deployment
- Educational cybersecurity project

9. Limitations

- User-space packet filtering
- Basic rule engine
- Limited protocol analysis

10. Future Enhancements

- GUI Dashboard
- Intrusion Detection Integration
- Machine Learning-based Anomaly Detection
- Real-time Alert System

11. Conclusion

The Personal Firewall successfully monitors and filters network traffic using Python and Scapy. The project demonstrates practical concepts of packet inspection, traffic monitoring, and network security.